PHYS-E0411 Advanced Physics Laboratory

Report: Superconducting Niobium Cavity

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2 Introduction

Superconductivity is a quantum mechanical phenomenon characterized by the full disappearance of electrical resistance below a material specific critical temperature T_C. This property is particularly valuable in applications that demand extremely low energy loss such as quantum information systems and particle accelerators [1]. Among elemental superconductors, niobium is notable for its relatively high T_C, making it suitable for experimental studies using liquid helium cooling.

In this experiment we investigated the microwave response of a three-dimensional rectangular niobium cavity operating in the superconducting regime. Such cavities support standing electromagnetic wave modes, with the TE_{101} mode being the fundamental resonance in our setup. The cavity was cooled to cryogenic temperatures using a helium-based cryostat and its response was probed using a Vector Network Analyzer via weak capacitive coupling. The key observable was the reflection coefficient S_{11} which provides insight into how much power is reflected back from the cavity as a function of frequency and temperature.

The goal of this experiment was to characterize the internal losses in the cavity and extract their temperature dependence. Using the S_{11} data we analyzed the resonator's response across a range of temperatures particularly focusing on the transition through T_C. We converted raw data into usable forms, performed curve fitting to Lorentzian lineshapes to extract internal damping rates κ_{int} and its variation with temperature. From this we aim to estimate the critical temperature T_C and evaluate the internal loss rate as a function of temperature.

3 Theory

3.1 Microwave Techniques

The frequency band of 1 GHz to 100 GHz in the electromagnetic spectrum belongs to microwaves. The wavelength ranges between the order of one meter down to a millimeter. There is a fundamental difference between the propagation of electromagnetic fields in electrical circuits at low or high frequencies. Electrical circuits contain wires, resistors, inductors, and capacitors, and can be described in terms of Impedance of each of these lumped elements. However, at higher frequencies, this description brakes down. The reason is that the fields have spatial dependence in the wire that is different than the dimensions of the connecting wires. In ordinary circuit analysis, both voltage and current may exhibit variations over the circuit's dimension in terms of magnitude and phase at higher frequencies. Instead of using a lumped element model to bridge the gap between field analysis and basic circuit theory analysis for microwave circuits and devices, a distributed element model of transmission line theory is employed.

3.1.1 Transmission Line Theory

One of the ways microwaves can propagate, is through guided structures like waveguides. The most common type of waveguide is the transmission line which is used to transmit electrical signals. Transmission line is made of at least two conductors. Two conductors placed near each other leads to

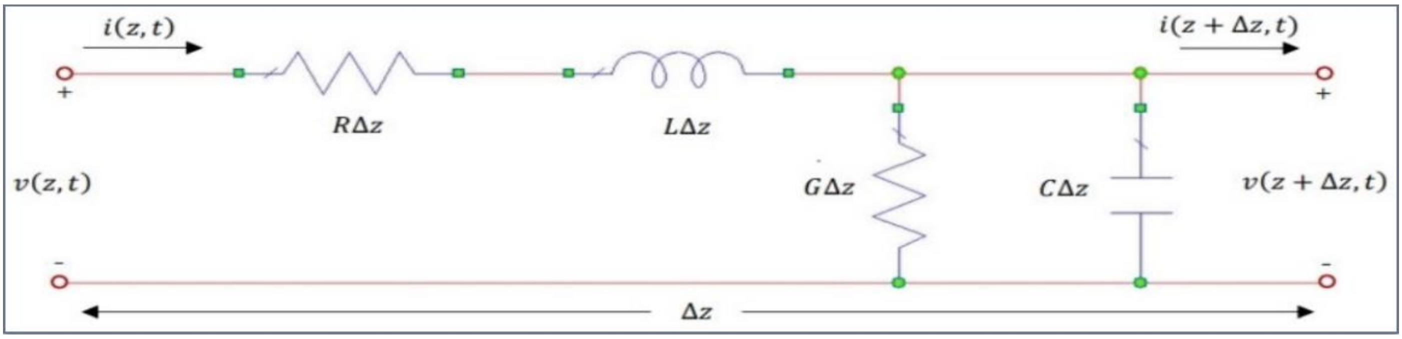


Figure 1: A section of discrete element model shown as an equivalent lumped element model

positive and negative charges accumulating that creates an electric field that stores energy (capacitive effect). A conductor carrying current creates a magnetic field that stores energy (inductive effect). In figure 1, for a transmission line the circuit elements L and C are due to self-inductance and capacitance between the conductors. R arises from finite conductivity of the conductors. G arises from dielectric loss between the conductors. For conductors with low loss properties, R and G are considered to be zero. By applying Kirchoff's Voltage law and Kirchoff's Current law in the equivalent model with approximation of $\Delta z \to 0$, we obtain

$$\frac{\partial v(z,t)}{\partial z} = -Ri(z,t) - L\frac{\partial i(z,t)}{\partial t} \tag{1}$$

$$\frac{\partial i(z,t)}{\partial z} = -Gv(z,t) - C\frac{\partial v(z,t)}{\partial t} \tag{2}$$

Since we deal with low loss device, we set R=G=0 and simplify the above first order equations (derivate both sides w.r.t z after substituting into each other) into separate second order equations

$$\frac{\partial^2 v(z,t)}{\partial z^2} = LC\frac{\partial^2 v(z,t)}{\partial t^2} \tag{3}$$

$$\frac{\partial^2 i(z,t)}{\partial z^2} = LC\frac{\partial^2 i(z,t)}{\partial t^2} \tag{4}$$

Therefore,

$$\frac{\partial^2 v(z,t)}{\partial t^2} - \frac{1}{LC}\frac{\partial^2 v(z,t)}{\partial z^2} = 0 \; and \; \frac{\partial^2 i(z,t)}{\partial t^2} - \frac{1}{LC}\frac{\partial^2 i(z,t)}{\partial z^2} = 0 \tag{5}$$

With $c^2 = 1/LC$, (c is speed of wave) we have

$$\frac{\partial^2 v(z,t)}{\partial t^2} - c^2\frac{\partial^2 v(z,t)}{\partial z^2} = 0 \; and \; \frac{\partial^2 i(z,t)}{\partial t^2} - c^2\frac{\partial^2 i(z,t)}{\partial z^2} = 0 \tag{6}$$

To note that R, L, G and C are all per unit length. Now with $L = 0.3\mu H/m$ and $C = 100pF/m$ we will have $c = \frac{1}{\sqrt{LC}} = 182574185.8\; m/s$ which is close to 66% of speed of electromagnetic wave (light).

Upon solving the equation by having $v = V(z)e^{i\omega t}$ *and* $i = I(z)e^{i\omega t}$, we get

$$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z} \tag{7}$$

$$I(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z} \tag{8}$$

$e^{-\gamma z}$ represents wave propagation in the positive z-direction, and $e^{\gamma z}$ represents wave propagation in the negative z-direction. The propagation constant is $\gamma = i\omega\sqrt{LC}$, where ω is the angular frequency of the travelling wave.

Characteristic Impedance is defined as the ratio of voltage and current waves in a transmission line. Therefore, the characteristic impedance of the transmission line in the loss less case is given by

$$Z_0 = \frac{V}{I} = \sqrt{\frac{L}{C}} \tag{9}$$

3.1.2 Electrical Resonators

Transmission lines can be used to make electrical resonators. A resonator is an electronic device that stores electromagnetic energy. A resonator 'resonates' at a particular electromagnetic wave frequency where a standing wave pattern is generated that causes the electromagnetic energy to be trapped and stored within the resonator. RLC circuits are used to represent resonators, which exhibit resonance at particular frequency. Using a parallel RLC circuit configuration, the Impedance when a voltage V is applied across the circuit can be expressed by

$$Z = \left(R + i\omega L + \frac{1}{i\omega C}\right)^{-1} \tag{10}$$

The average magnetic energy stored in the inductor: $\frac{1}{4}|I|^2 L$

The average electric energy stored in the capacitor: $\frac{1}{4}|V|^2 C = \frac{1}{4}\frac{|I|^2}{\omega^2 C}$

Resonance occurs when average magnetic energy is equal to average electric energy. The impedance will be real valued. The resonance frequency is then given by

$$\omega_0 = \frac{1}{\sqrt{LC}} = 2\pi f_0 \tag{11}$$

3.1.3 Quality Factor

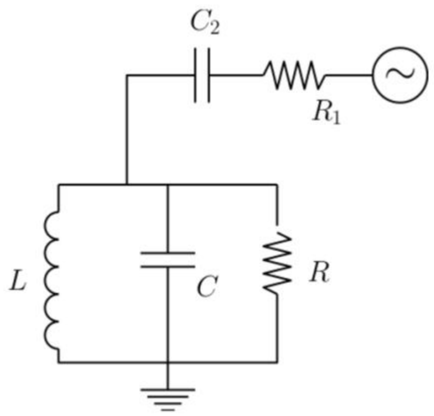


Figure 2: Parallel RLC circuit capacitively coupled to external circuit

The quality factor Q quantifies how well the resonator stores energy relative to how quickly it loses it. It is defined as

$$Q = \frac{2\pi E}{\Delta E} = \frac{f_0}{\kappa} \tag{12}$$

with κ being the energy damping rate. For a parallel RLC resonator, the quality factor in terms of its circuit elements is $Q = R/\omega_0 L = R\omega_0 C$.

In this experiment, the resonator is connected to the external circuit through a capacitive coupling configuration as shown in figure 2. This is done to ensure weak but controlled coupling, preserving a high quality factor while still allowing signal readout. The coupling is realized through a small capacitor C_c, which effectively introduces an external dissipation resistance:

$$R_{ext} \approx \frac{1}{\omega_0^2 C_c^2 Z_0} \tag{13}$$

When the resonator is connected to the external world, both internal and external losses contribute to the total damping. These are captured through the internal and external quality factors:

$$Q_{ext} = \frac{R_{ext}}{\omega_0 L} = R_{ext}\omega_0 C \tag{14}$$

$$Q_{int} = \frac{R_{int}}{\omega_0 L} = R_{int}\omega_0 C \tag{15}$$

The combined or loaded quality factor is given by $Q^{-1} = Q_{ext}^{-1} + Q_{int}^{-1}$ and the total energy damping rate is simply $\kappa = \kappa_{ext} + \kappa_{int}$. To achieve optimal measurement conditions without compromising the resonator's intrinsic performance the coupling strength is tuned such that $\kappa_{ext} \approx \kappa_{int}$. This provides a balance between readable output and preserved resonance behavior. Close to resonance the frequency response follows a Lorentzian profile. The reflected signal magnitude is given by:

$$(C_k)^2 |Z|^2 \approx \frac{\left(\frac{\kappa}{2}\right)^2}{(\omega - \omega_0)^2 + \left(\frac{\kappa}{2}\right)^2} \tag{16}$$

This describes a sharp resonance peak at ω_0 where the energy storage and reflection behavior are dominated by the total loss rate κ.

3.1.4 Rectangular microwave cavity

Rectangular microwave cavities are commonly used to confine electromagnetic energy in three dimensions by forming closed metallic enclosures. These cavities act as resonators by supporting standing wave patterns of the electric and magnetic fields. Such confinement gives rise to discrete resonant frequencies that depend directly on the physical dimensions of the cavity.

In this experiment, we study a rectangular 3D cavity resonator made of superconducting niobium. At microwave frequencies such cavities support transverse electric (TE) or transverse magnetic (TM) modes. The field configuration associated with each mode is described by integers m, n, l which represent the number of half-wavelength variations along the x, y, z dimensions, respectively. The resonant frequency for a given TE_{mnl} or TM_{mnl} mode is given by

$$f_{mnl} = \frac{c}{2\pi\sqrt{\mu_r\epsilon_r}}\sqrt{\left(\frac{m\pi}{X}\right)^2 + \left(\frac{n\pi}{Y}\right)^2 + \left(\frac{l\pi}{Z}\right)^2} \tag{17}$$

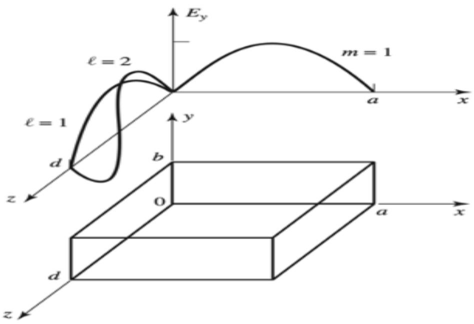


Figure 3: A rectangular cavity resonator, and the electric field variations for the TE_{101} and TE_{102} resonant modes.

Here X, Y, Z are the cavity dimensions, μ_r and ϵ_r are the relative permeability and permittivity of the cavity medium (typically vacuum), and c is the speed of light in vacuum. In our specific geometry where $Y < X < Z$ the lowest resonant mode is the TE_{101} mode, which is the fundamental mode used in this study. This mode has the electric field varying along x and z, but uniform in y, making it suitable for efficient excitation and detection via coupling pins aligned with the field.

3.1.5 Reflection Measurement

To characterize how a microwave cavity behaves across different frequencies and what happens at resonance frequency, we make use of a quantity known as the reflection coefficient S_{11}. This parameter tells us how much of an incident electromagnetic wave is reflected back when it encounters a device or discontinuity in the transmission line. Mathematically, if an incoming wave traveling to the right is denoted as $V^-(z)$ then

$$\frac{V^-(z)}{V^+(z)} = S_{11} \tag{18}$$

where $V^{\pm}(z) = V_0^{\pm}e^{\mp\gamma z}$. Here S_{11} is the complex reflection coefficient that depends on how different the impedance of the device (Z) is from the characteristic impedance of the transmission line (Z_0). This mismatch causes part of the wave to be reflected and the reflection coefficient is defined by

$$S_{11} = \frac{Z - Z_0}{Z + Z_0} \tag{19}$$

In this experiment our device under test is the niobium cavity and we measure response using a Vector Network Analyzer. The VNA sends a signal into the system and measures the reflected signal's amplitude and phase, allowing us to identify resonance behavior. At resonance the cavity absorbs energy efficiently and the reflected signal drops giving rise to a dip in $|S_{11}|$. This frequency dependent behavior of $|S_{11}|$ near the resonance frequency follows a Lorentzian profile:

$$|S_{11}|^2 = \frac{(\omega - \omega_0)^2 + \left(\frac{\kappa_{ext} - \kappa_{int}}{2}\right)^2}{(\omega - \omega_0)^2 + \left(\frac{\kappa}{2}\right)^2} \tag{20}$$

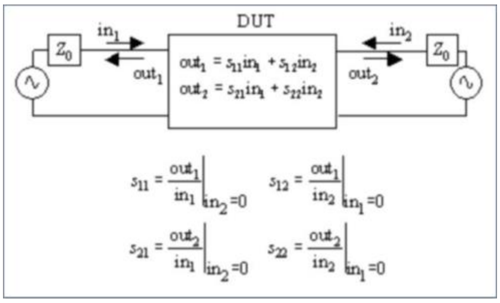


Figure 4: S-Parameter Representation

where $\kappa = \kappa_{ext} + \kappa_{int}$ is the total damping rate of the cavity, combining both external (coupling-related) and internal (material-related) losses rate. The minimum of this curve indicates the resonance frequency and its width gives insight into the energy loss. This process is best described using the framework of scattering parameters. In a two-port system as shown in figure 4, each S-parameter describes the ratio of outgoing to incoming waves at a given port, under specific conditions. For example, S_{11} describes reflection at port 1 and S_{21} describes transmission from port 1 to port 2.

These parameters are complex quantities with both magnitude and phase components. The magnitude is typically expressed in decibels (dB) and for S_{11} it is written as $20 \log |S_{11}|$. This is also the ratio of measured signal power to a reference signal power. At resonance frequency we see a dip in the s-parameter spectrum. The phase carries information about the wave delay or advancement useful for full system characterization.

In this experiment by analyzing the reflection coefficient S_{11} over a range of frequencies, we are able to identify the resonance characteristics of the cavity and extract critical parameters such as the quality factor and damping rates [2].

3.2 Low Temperature Techniques

3.2.1 Superconductivity

The first macroscopic quantum phenomenon that was discovered experimentally was Superconductivity [3]. This is a unique quantum phenomenon observed in certain materials at cryogenic temperatures, where they exhibit zero electrical resistance and perfect diamagnetism. These materials transition into a superconducting state below a specific temperature called the critical temperature T_C. The earliest discovery was made by Heike Kamerlingh Onnes in 1911 in mercury [4], and a robust microscopic theory, known as BCS theory, was later developed in 1957 by Bardeen, Cooper, and Schrieffer.

The foundation of this behavior lies in the formation of Cooper pairs: paired electrons that move without scattering, thus eliminating resistance under direct current (DC) conditions [5]. Alongside zero resistance, superconductors exhibit the Meissner effect, which is the expulsion of magnetic fields from their interior when cooled below T_C. A key quantum property of superconductors is the existence of an energy gap Δ_0 near the Fermi level, which prevents single-particle excitations below the critical temperature. This gap is related to T_C as

$$\Delta_0 \approx 1.76 \frac{k_B T_C}{e} \tag{21}$$

Here, k_B is Boltzmann's constant and e is the elementary charge [6]. The gap is typically in the range of a few millielectronvolts. While superconductors conduct perfectly in DC, their behavior in alternating current (AC) or microwave fields is more nuanced. At high frequencies there is a small but measurable resistance due to the inertia of the Cooper pairs predicted by London in 1934. This was shown experimentally when at transition temperature there was a big change in conductivity observed at microwave frequencies around 1.5 GHz [7]. Since there was no change in optical absorption during the superconducting transition, it became clear that conductivity as a function of frequency showed a change from a superconducting state to a low loss metal state at microwave frequencies [8]. As a result, AC conductivity becomes complex and frequency-dependent. The frequency response can be modeled by

$$\sigma(\omega) = \frac{\sigma_{DC}}{1 + i\omega\tau} \tag{22}$$

where ω is the angular frequency and τ is the electron scattering time. In the superconducting state, τ becomes very large, and the AC conductivity becomes largely inductive in nature. Because the microscopic electric field does not penetrate deeply into the superconductor, an alternative surface-based measure is used which is expressed as

$$Z_s = R_s + iX_s \tag{23}$$

where R_s is the surface resistance, and $X_s = \omega L_s$ represents the surface reactance due to inductance. Although inductive contributions like kinetic inductance L_k play a role in some systems (especially thin films like twisted bilayer graphene), they are secondary in our bulk 3D cavity system [9].

3.2.2 Losses in Superconductors

Even though superconductors ideally exhibit zero resistance, in practical high frequency applications such as microwave cavities, losses still occur due to residual surface effects. These are especially important in the evaluation of resonator quality factors and device performance in quantum technologies. The primary mechanism of loss in both normal and superconducting cavities is related to surface resistance R_s. At high frequencies current flows are restricted to a thin layer near the conductor surface known as the skin effect. The effective resistance increases because of the very limited depth to which AC fields can penetrate. This surface resistance is related to the material's bulk resistivity ρ and skin depth δ_s as

$$R_s = \frac{\rho}{\delta_s} \tag{24}$$

For superconductors, this resistance and its temperature dependence can be modeled by:

$$R_s \approx \alpha \left(\frac{T}{T_c}\right)^4 \left[1 - \left(\frac{T}{T_c}\right)^4\right]^{-\frac{3}{2}} + R_0 \tag{25}$$

Here, α is a constant, and R_0 accounts for additional residual losses such as dielectric losses. In resonant cavities, the impact of surface resistance is captured through the internal quality factor Q_{int}, which quantifies how many oscillations the system sustains before the energy is dissipated. For a rectangular 3D cavity with dimensions X, Y, Z the quality factor is given by:

$$Q_{int} = \frac{(kXZ)^3 Y \eta_0}{2\pi^2 R_s} \frac{1}{2X^3Y + 2YZ^3 + X^3Z + XZ^3} \tag{26}$$

where, $k = \omega_0\sqrt{\mu_0\epsilon_0}$ is the wave number, $\eta_0 = \sqrt{\mu_0/\epsilon_0} = 377\Omega$ is the free space impedance and ω_0 is the resonant angular frequency. These expressions help link the material properties of superconductors to measurable quantities like resonance sharpness and energy loss. In this experiment, extracting Q_{int} from reflection measurements allows us to evaluate how resistive losses evolve with temperature and estimate the superconducting transition.

3.2.3 Niobium as a superconducting material

Niobium as an intrinsic type-I superconductor but with impurities it is a type-II superconductor [10]. Pure Niobium is two active bands and two gap superconductor. The bands are moderately anisotropic with temperature dependent anisotropies. The more isotropic band 2 dominates the electronic properties. The superconducting gap is between 1.55-1.43 meV [11]. Some other literature values also state it to be 2.32 meV [12].

Niobium is a very important element which has use cases in steel production, superalloys, electroceramics, magnetic resonance imaging, nuclear magnetic resonance etc. It's most important application is in superconducting microwave/RF devices. Pure niobium is utilised to create the superconducting radio frequency (SRF) cavities used in the FLASH and XFEL free-electron lasers. The International Linear Collider's 30-kilometer linear particle accelerator will use 1.3 GHz nine-cell SRF cavities created by a cryomodule team at Fermilab [13]. Superconducting niobium nitride bolometers are a perfect detector for electromagnetic radiation in the THz frequency range due to their great sensitivity. The HIFI instrument aboard the Herschel Space Observatory uses these detectors, which were tested at the Submillimeter Telescope, the South Pole Telescope, the Receiver Lab Telescope, and at APEX [14].

4 Measurements

4.1 Equipment

Cryostat (with vacuum can): The cryostat is a tall, insulated structure used to house and cool the superconducting cavity. It is fitted with a vacuum can that prevents boiling cryogens from coming into direct contact with the sample. The sample cavity is attached to the copper cold finger at the bottom which provides efficient thermal conduction from the cryoliquid bath to the cavity.

Copper cold finger: This is the component that physically connects the cavity to the cryostat and conducts heat away from the sample. It ensures the cavity reaches and maintains the low temperatures needed for superconductivity.

DT-670 silicon diode thermometer: Mounted near the cavity this temperature sensor provides voltage readings which are converted into accurate temperature values. It is sensitive enough to monitor changes down to just a few Kelvin.

Vector Network Analyzer: The VNA is used to probe the cavity using microwave signals. It sends a signal into the system and measures how much is reflected back. This gives us the reflection coefficient $|S_{11}|$ from which the resonant frequency and quality factor of the cavity can be extracted.

Directional coupler: This device separates the incoming and reflected signals, allowing the VNA to correctly interpret the reflected signal from the cavity, even though the measurement is formally taken at port 2 (simulating $|S_{11}|\ as\ |S_{21}|$)

SMA cables and connectors: These are the microwave cables that connect the VNA to the cavity and ensure good signal integrity. They are tightened with a torque wrench to guarantee consistent contact.

Coupling pin: This pin attached to the cavity via an SMA feedthrough injects the probing microwave signal into the electric field of the cavity. Its positioning ensures strong enough coupling for measurement without affecting the resonator.

Liquid nitrogen dewar: A smaller container used in the pre-cooling stage. The cryostat is first dipped into this to bring the cavity temperature down to around 100 K before switching to helium for deeper cooling.

Liquid helium dewar: A large insulated vessel used to cool the system below 4.2 K, essential for niobium to reach its superconducting state. Once pre-cooled the cryostat is submerged in this dewar to reach cryogenic operating temperatures.

Indium wire and vacuum grease: These materials are used to ensure a good vacuum seal inside the cryostat. The indium is pressed into place and the vacuum grease improves the sealing surface contact.

Vacuum pump system: Connected to the vacuum can, this removes air and residual gases before introducing the cryogen. It helps reduce thermal conduction and ensures better cooling efficiency.

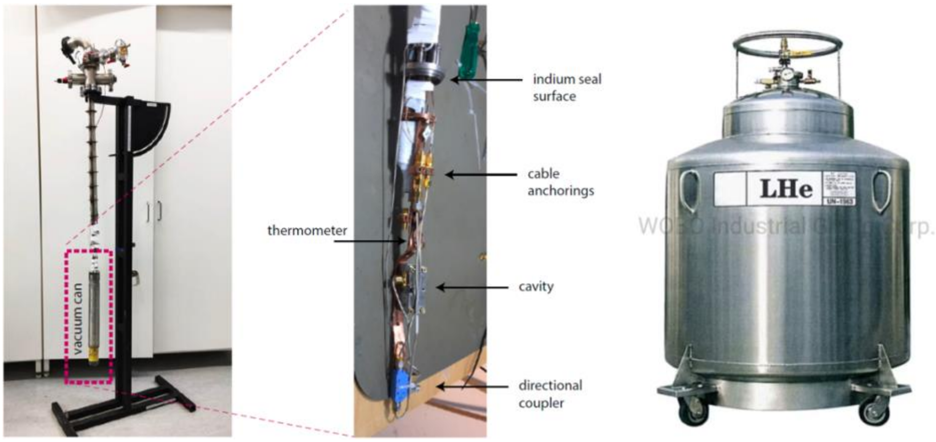


Figure 5: left – cryostat with vacuum can and its component, right – liquid helium dewar

4.2 Methods

Initial Setup: The experiment starts with basic safety preparations wearing protective gear such as rubber gloves or leather gloves (face shield as well for the person operating). The cavity is first inspected and cleaned using acetone to remove any contaminants. Its internal dimensions are carefully measured with a vernier caliper to later compare theoretical resonance frequencies with the experimental result.

Assembly and Cabling: The cavity is closed using screws and nuts ensuring proper alignment. It is then attached to the VNA using an SMA connector. The connection is made through a coupling pin feedthrough allowing the microwave signal to excite the cavity's electric field. The VNA is configured to sweep around the expected resonance frequency (approximately 11.37 GHz), and the actual resonance peak is found with a span of ± 2.5 MHz and 0.1 MHz accuracy.

Cryogenic Part: After initial signal testing the cavity is disconnected from the VNA and mounted onto the copper cold finger of the cryostat. An SMA cable is reconnected to the cavity, and the DT-670 thermometer is checked for proper operation. A cleaned indium wire is prepared and inserted into the vacuum can's sealing surface coated with vacuum grease to ensure a tight vacuum seal.

Vacuum Sealing and Pre-cooling: The vacuum can is positioned over the cold finger and sealed. A vacuum pump is connected and run for several minutes to evacuate the can. Pre-cooling is carried out by lifting the cryostat into a liquid nitrogen dewar, submerging the vacuum can partially. Cooling continues until the thermometer reads approximately 100 K.

Transition to Liquid Helium: Once pre-cooled, the cryostat is lifted from the nitrogen bath and submerged into a liquid helium dewar. This lowers the temperature further to close to 4.2 K, necessary for niobium to enter its superconducting state. Caution is observed during this stage due to high-speed helium gas jets from the dewar.

Measurement: When the system stabilizes below 20 K, the reflection measurement script is initiated from a pre-written Matlab interface. The VNA records the reflection coefficient $|S_{11}|$ over a set frequency span. As the temperature crosses the superconducting transition significant changes in the resonance dip and Q-factor (dip width and depth) are observed and recorded. The measurement is repeated across many temperature points particularly focusing on the range near the transition temperature.

Data Collection: Voltage values from the silicon diode are logged for each scan. These are later converted into temperature values using a given calibration function or dataset. The corresponding resonance curves (in $|S_{11}|$ vs frequency) are stored for further analysis.

5 Analysis and Results

Frequency of TE_{101} mode: Using equation 17, we can use the measured cavity dimensions (using a vernier caliper) to find the frequency. Also since the μ_r and ϵ_r depend on the medium inside cavity, we take these as 1 (for air).

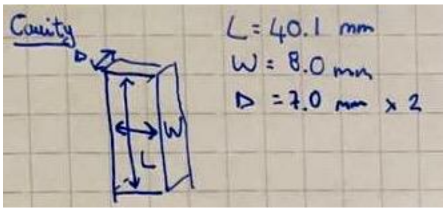


Figure 6: Cavity dimension values

$$\therefore f_{101} = \frac{3 \times 10^8}{2\pi\sqrt{1}} \sqrt{\left(\frac{1\pi}{14 \times 10^{-3}}\right)^2 + \left(\frac{0\pi}{8 \times 10^{-3}}\right)^2 + \left(\frac{1\pi}{40.1 \times 10^{-3}}\right)^2} \approx 11.3485\ GHz$$

The stated value of the frequency is 11.37 GHz and the frequency found in the lab at room temperature was 11.431 GHz. These values are quite close to the one calculated above. All these values might be slightly different because the cavity used in the lab was not exactly cuboidal and had semicircular ends. This would slightly change the boundary conditions of the differential equations that must be solved for the electromagnetic fields to extract the frequency. The dimension values also depend on the accuracy of the vernier caliper. We can use simulation tools like COMSOL to predict the frequency of any cavity structure if we can apply the correct boundary physics and have the right geometry [15].

Voltage to Temperature conversion: Using the data recorded during experiment and the TempCal.m Matlab function given in MyCourses, the variables were loaded in the workspace and the TempCal function called on the array of $-1.0 \times$ 'Voltage' which consists of the thermometer voltage readings. This data / new array was saved and now loaded in Python to display and use it further, as shown in figure 7.

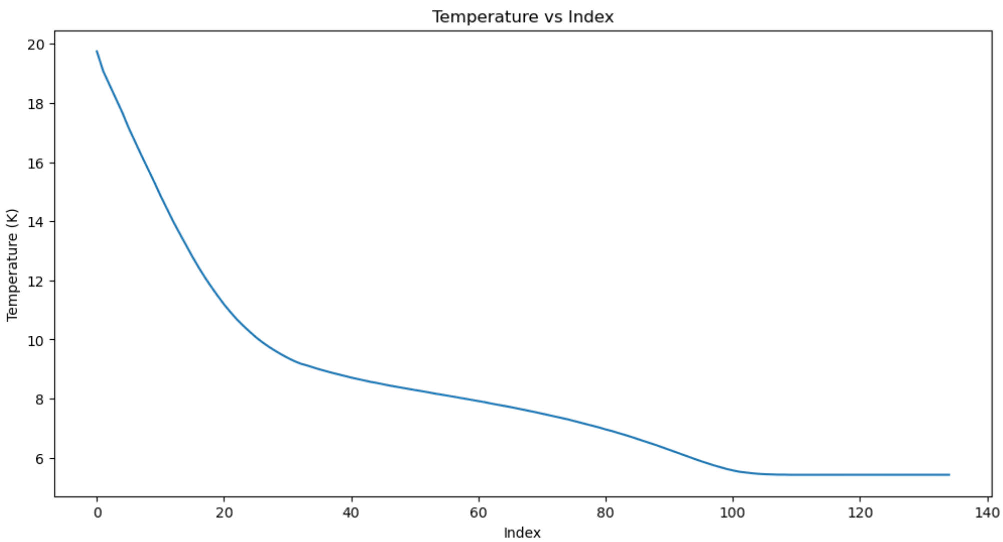


Figure 7: Temperature values for 135 indices

Resonance curve fitting to obtain parameters: The data acquired was $|S_{11}|$ dB vs frequency (f) for a range of temperature values. The idea here is to fit this data according to equation 20 and extract unknown parameters, for each temperature / index. According to equation 20, we know $|S_{11}|^2$ (converted from $10^{\frac{S_{11,dB}}{10}}$), $\kappa_{ext} = 10^7\ Hz$ and angular frequencies $\omega = 2\pi f$. What is not known is κ_{int} and resonant angular frequency $\omega_0 = 2\pi f_0$. The fitting is done in python and the model function used is equation 20. I first fit the model to the entire frequency range. The results were unreliable: many fits failed or returned unrealistic κ_{int} values. This happened because most data points lie on a flat baseline; in a least-squares fit those baseline points outweigh the few points inside the resonance dip (out of bounds as the interpreter raises a run-time error), and so it focuses on the background instead of the dip. To locate the useful part of the spectrum I plotted $|S_{11}|^2$ versus frequency for several temperatures as seen in figure 8 and saw that every dip falls between frequency indices 350 and 550. Restricting the fit to this 350-550 window keeps all points that contain information about the dip's depth and width while excluding the flat baseline. This greatly improves convergence and yields consistent physically meaningful κ_{int} values. The fits and data for various temperatures and the analysis code can be seen in the appendix section.

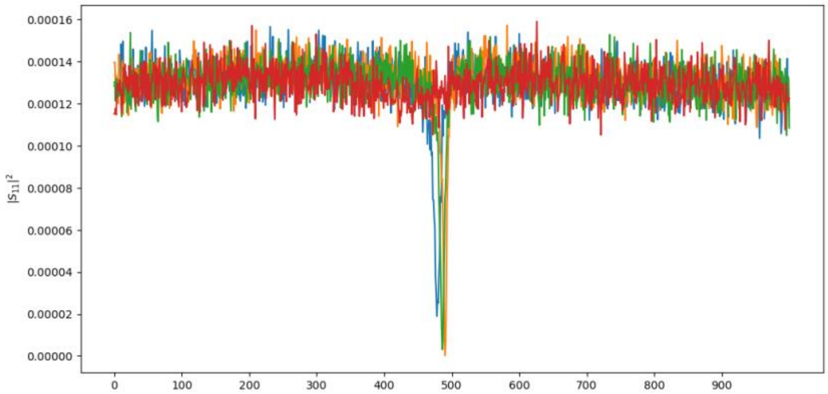


Figure 8: Test $|S_{11}|^2$ vs frequency plots to understand which range dip lies in

Internal loss rate vs Temperature: By fitting $|S_{11}|^2$ vs frequency for each temperature value, thereby we can extract κ_{int} for each temperature value. This is plotted in figure 9, we can see that the transition happens somewhere between 8 and 10 K hence the transition temperature must lie in this range.

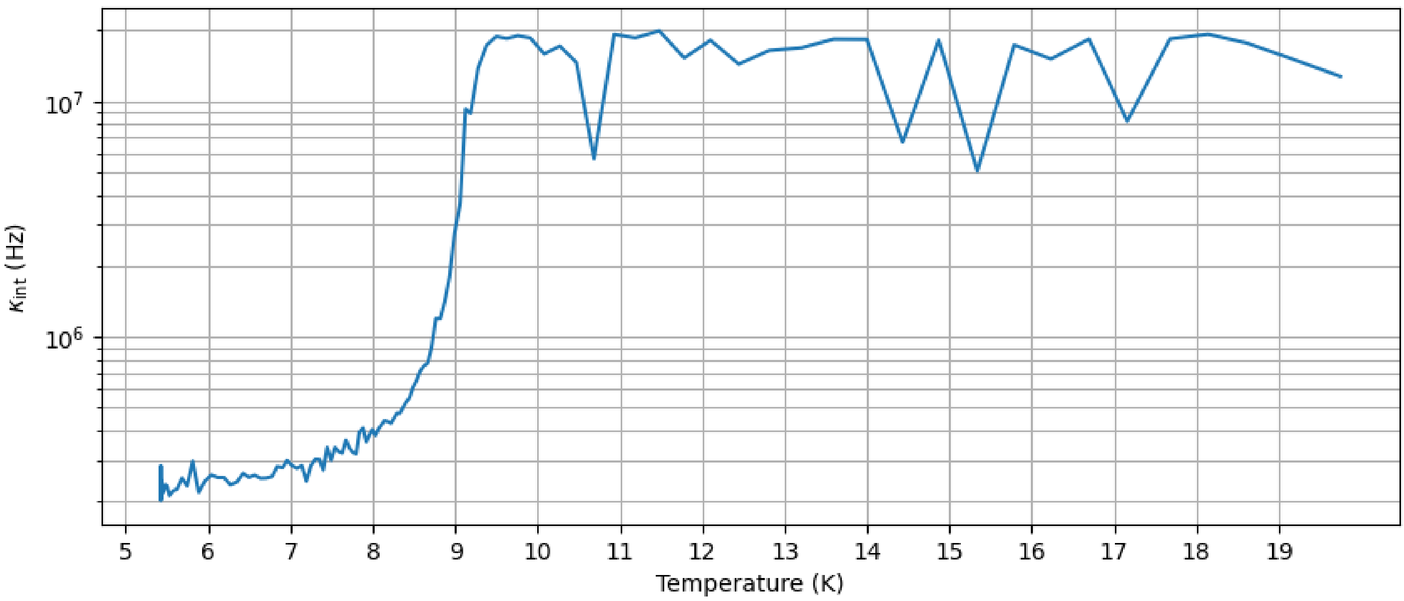


Figure 9: Internal loss rate vs Temperature plot

Fitting temperature dependence of κ_{int}: First we use equation 21 with $\Delta_0 \approx 0.002$ eV (guess in fitting) which will be one of the fitting parameter and substitute T_c in equation 25. Here we take $\alpha = 10^{-3}$ *and* $R_0 = 10^{-3}\Omega$ (guesses) and substitute R_s in equation 26. Equation 26 will now take the form-

$$\frac{f_0}{\kappa_{int}} = \frac{k^3X^3Z^3Y\eta_0}{2\pi^2}\frac{1}{2X^3Y + 2YZ^3 + X^3Z + XZ^3}\frac{1}{R_s(T)}$$

$$\kappa_{int} = \frac{2\pi^2 f_0(2X^3Y + 2YZ^3 + X^3Z + XZ^3)}{8\pi^3 f_0^3 \epsilon_0^{3/2}\mu_0^{3/2}X^3Z^3Y\sqrt{\mu_0/\epsilon_0}}\frac{1}{R_s(T)}$$

$$\kappa_{int} = \frac{(2X^3Y + 2YZ^3 + X^3Z + XZ^3)}{4\pi f_0^2\epsilon_0\mu_0^2X^3Z^3Y}\frac{1}{R_s(T)} \tag{27}$$

We have κ_{int} *vs* T curve from figure 9 and will use equation 27 as the model function. X, Y, Z, f_0, ϵ_0, μ_0 are known values and extract fitting values of α, R_0 and mainly Δ_0 to find our critical temperature using equation 21. Also, since the equation works only in superconducting region we must limit the region in we conduct our fit. We take three such similar regions as seen in figure 10 and take the average value to maintain accuracy.

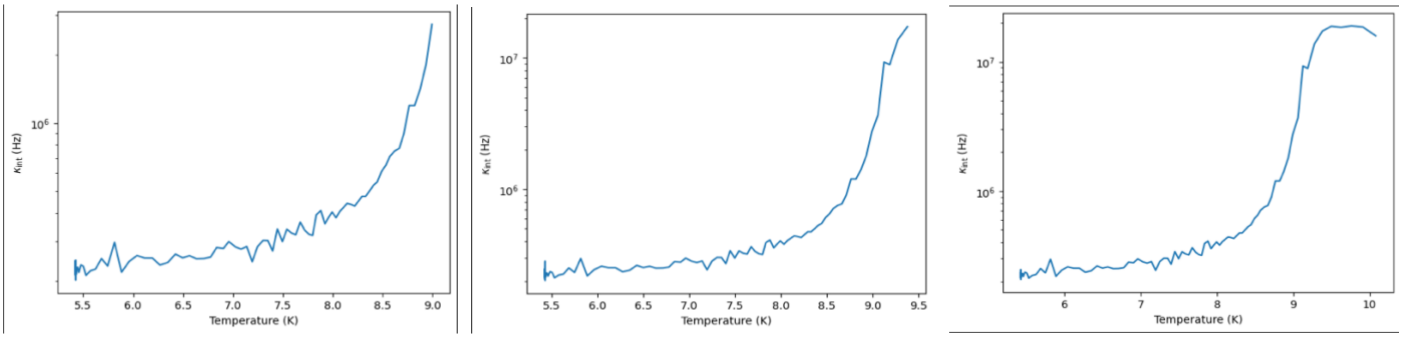
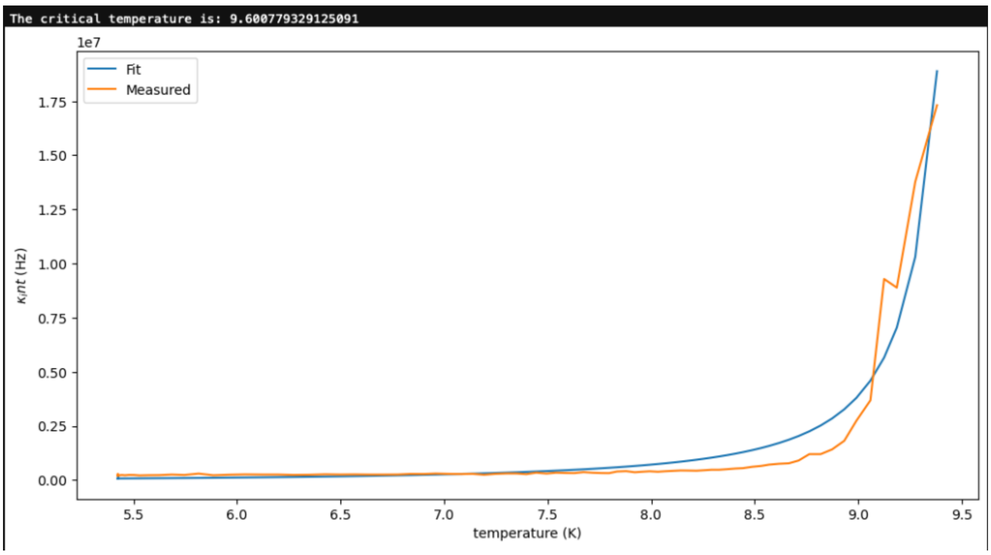
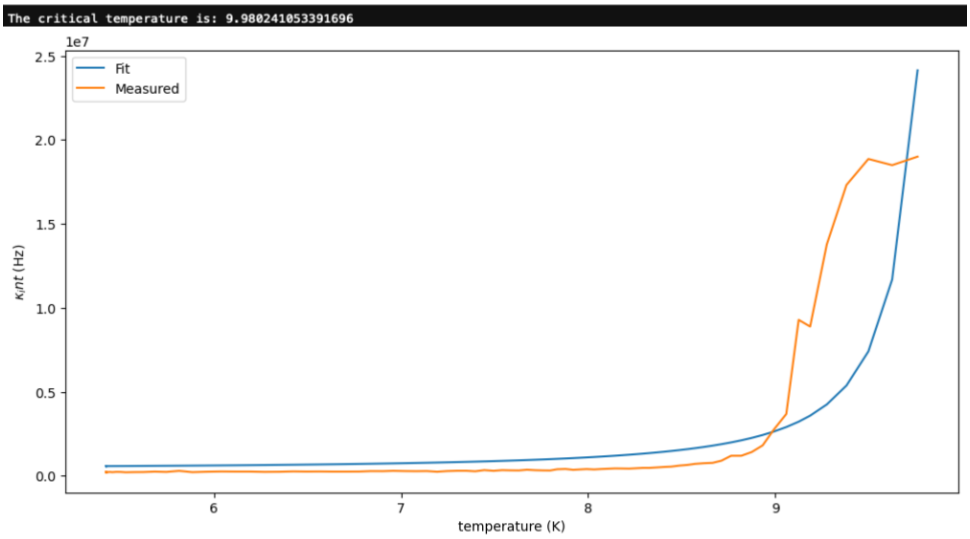


Figure 10: Internal loss rate vs Temperature for three similar regions (just shifted indices)

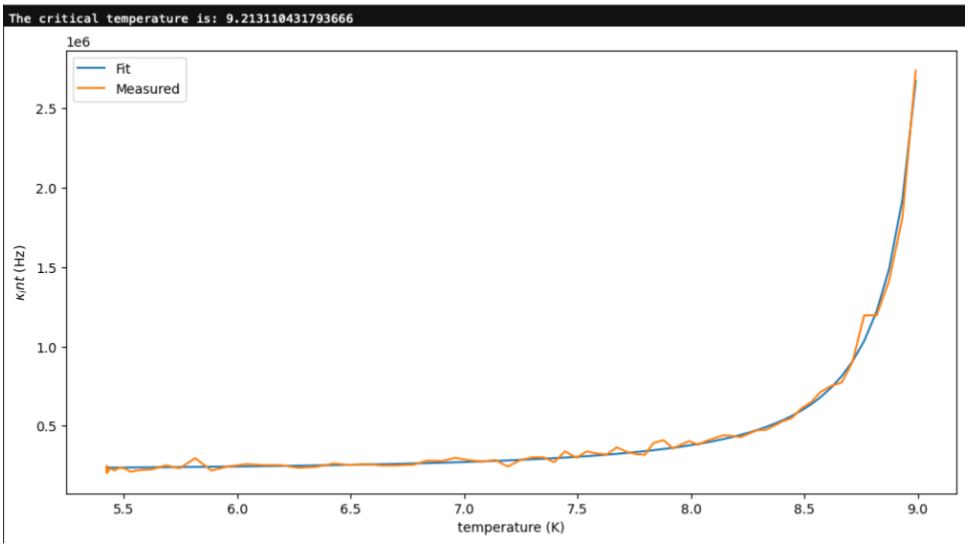
Determining Critical Temperature: By extracting Δ_0 from fitting for each region we can calculate critical temperature using equation 21. Literature states that the critical temperature of niobium is around 9.7 ± 0.6 K and previously known values also include 9.2 K [16]. From our fits shown in figure 11 we find that critical temperatures extracted from κ_{int} vs Temperature curve are 9.60 K, 9.98 K and 9.21 K. The average of these three values (rounded off to two decimal places) is 9.60 K which is very close to and in the range of the stated literature values.



(a)



(b)



(c)

Figure 11: Fitting of internal loss rate vs temperature to extract critical temperature

6 Conclusion

In this experiment the microwave response of a superconducting niobium 3D cavity resonator was studied by performing reflection measurements across a temperature range spanning the superconducting transition. Through careful analysis of the measured reflection coefficient $|S_{11}|^2$, we extracted key physical parameters of the cavity like the resonance frequency and most notably the internal loss rate κ_{int} and its dependence on temperature.

The resonance frequency obtained from electromagnetic mode analysis using the cavity's dimensions was calculated to be approximately 11.3485 GHz which showed good agreement with the 11.45 GHz resonance extracted from experimental fitting. This validates both our theoretical understanding of cavity modes and the quality of the measurement data. By fitting the measured data to the Lorentzian model we successfully extracted κ_{int} for each temperature. The plot of internal loss rate versus temperature revealed a clear superconducting transition. From the behavior of the internal losses across three regions and estimated the critical temperature T_c of the niobium cavity to be approximately 9.60 K which lies well within the expected range reported in literature texts.

Beyond data analysis this exercise also provided visual demonstration of working with cryogenic and microwave equipment such as vector network analyzer, directional coupler, and temperature control system and understanding how to set up the VNA to measure S_{11} through a directional coupling path, proper sealing and how the cryostat maintains ultra-low temperatures using liquid helium and nitrogen. Overall this lab successfully bridged theory and experiment from actual physical realization of electromagnetic resonance to data interpretation to extract useful information about the system's properties.

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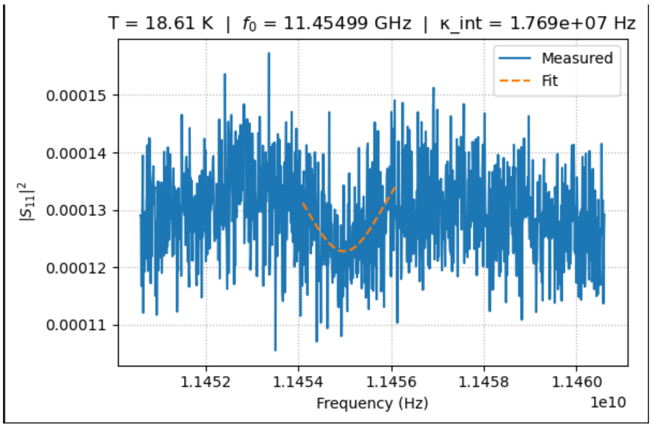
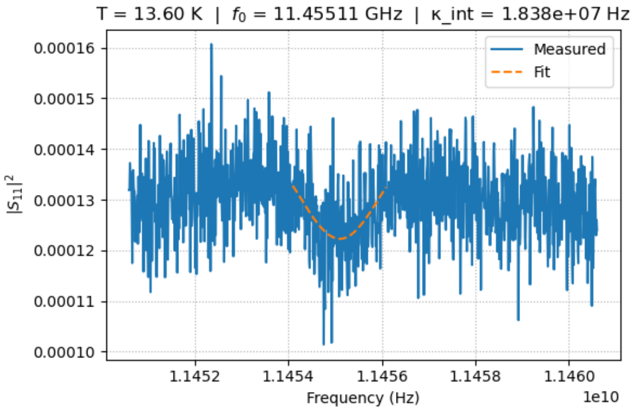
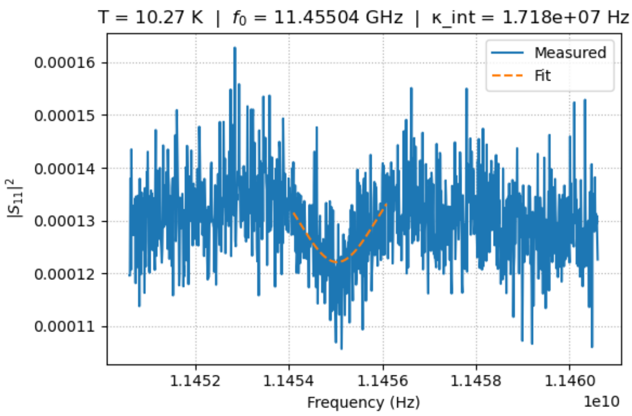
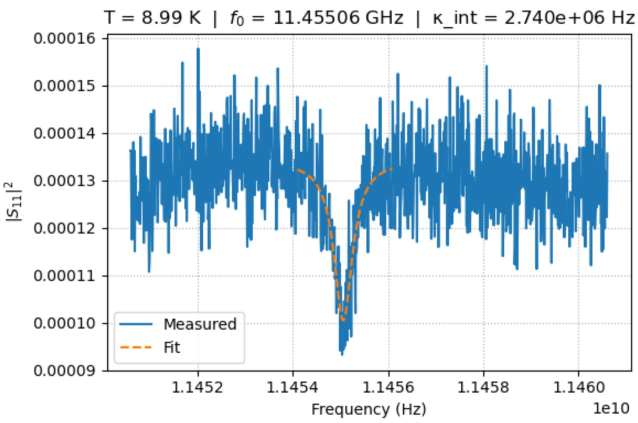
8 Appendix

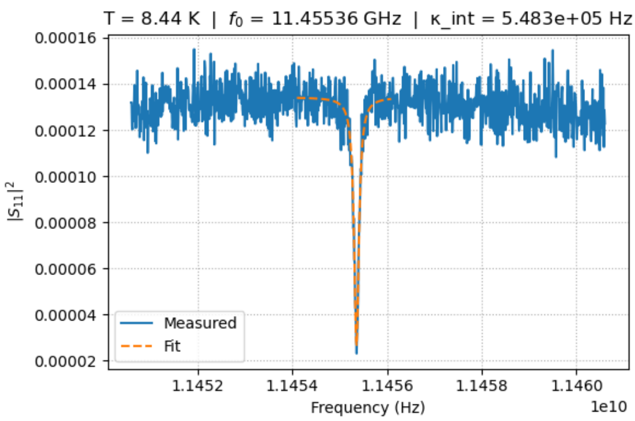
8.1 Resonance curves for different temperature values

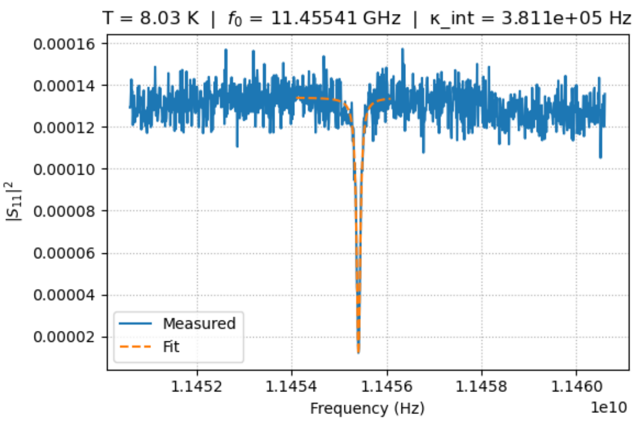


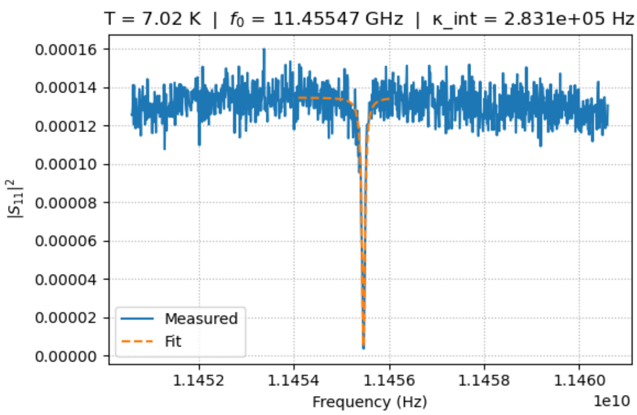


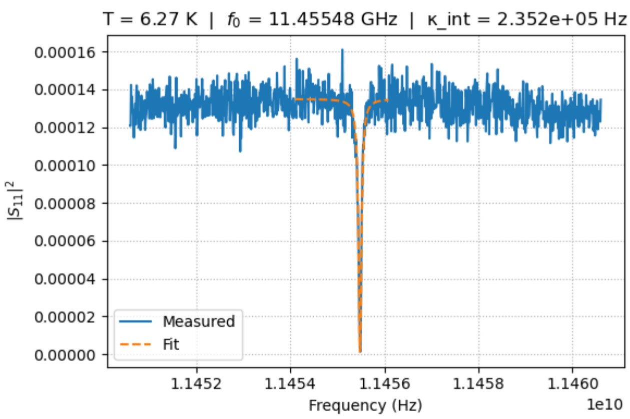












8.2 Code used during analysis

Importing data and fit function:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
from scipy.io import loadmat
```

```
data1=loadmat('2025-03-31-1559.mat')
data2=loadmat('TemperatureDatta.mat')
freq=np.squeeze(data1['dfreq'])
S11_mag_lin=10**(data1['datamag']/10.0)
Temperature=np.squeeze(data2['Temperature'])

def lorentzian(f, f0, k_int, k_ext, A):
    return A * ((2*np.pi*(f-f0))**2 + ((k_ext-k_int)/2.0)**2) / ((2*np.pi*(f-f0))**2 + ((k_ext+k_int)/2.0)**2)
```

Fitting for extracting internal loss rate:

```
p0=[11.461e9,1e7,1e7,1e-4]  #[f0,k_int,k_ext,A]
bounds=([11e9,1e4,1e4,0.0],[12e9,1e8,1e7,3e-4])
freq_fit=freq[slice(350,550)]

N=S11_mag_lin.shape[0]
f0_tot=np.empty(N)
k_int_tot=np.empty(N)
k_ext_tot=np.empty(N)
A_tot=np.empty(N)

for i in range(N):
    y=S11_mag_lin[i,slice(350,550)]
    popt, _=curve_fit(lorentzian,freq_fit,y,p0=p0,bounds=bounds,maxfev=10000)
    f0_tot[i],k_int_tot[i],k_ext_tot[i],A_tot[i] = popt
```

Resonance curves plotting:

```
idx=#100
plt.plot(freq,S11_mag_lin[idx],label='Measured')
plt.plot(freq_fit,lorentzian(freq_fit,f0_tot[idx],k_int_tot[idx],k_ext_tot[idx],A_tot[idx]),'--',label='Fit')
plt.xlabel('Frequency (Hz)')
plt.ylabel(r'$|S_{11}|^{2}$')
plt.title(f'T = {Temperature[idx]:.2f} K  |  '
    fr'$f_0$ = {f0_tot[idx]/1e9:.5f} GHz  |  '
    fr'k_int = {k_int_tot[idx]:.3e} Hz')
plt.legend()
plt.grid(ls=':')
```

Interna loss rate vs temperature plot:

```
plt.plot(Temperature,k_int_tot)
plt.yscale('log')
plt.xlabel('Temperature (K)')
plt.ylabel(r'$\kappa_{\rm int}$ (Hz)')
plt.xticks(np.arange(5,20,step=1))
plt.grid(True,which='both')
```

Internal loss rate theory relation:

```
X=14.0e-3
Y=8.0e-3
Z=40.1e-3
e=1.6022e-19
k_b=1.38e-23
mu_0=1.2566*1e-6
eps_0=8.854e-12
f_0=np.mean(f0_tot)


def test(T,alpha,scg,residue):
    a=(alpha*(T/(scg*e/(1.76*k_b)))**4) * ((abs(1-(T/(del0*e/(1.76 * k_b)))**4))**(-3/2)) +residue
    return a * (2*Y * Z**3 + 2*Y * X**3 + Z * X**3 + X * Z**3)/(X**3 * Z**3 * Y * mu_0*mu_0*eps_0*np.pi*4 * f_0**2)


alpha,scg,residue=1e-3,0.002,1e-3 #guess
bounds=([1e-4, 0.001, 1e-4],[1e-2, 0.002, 1e-2])
```

Critical temperature extraction:

```
x=Temperature[35:125]
y=k_int_tot[35:125]
u,v=curve_fit(test,x,y,p0=[alpha,scg,residue],maxfev=10000,bounds=bounds)
output=[test(i,u[0],u[1],u[2]) for i in x]
plt.plot(x,output,label='Fit')
plt.plot(x,y,label='Measured')
plt.xlabel('temperature (K)')
plt.ylabel(r'$\kappa_int$ (Hz)')
crit=u[1]*e/(1.76*k_b)
print('The critical temperature is:', crit)
```